Practice problems #14

1. Let $u_1 = \begin{bmatrix} 0 \\ 1 \\ -4 \\ -1 \end{bmatrix}$, $u_2 = \begin{bmatrix} 3 \\ 5 \\ 1 \\ 1 \end{bmatrix}$, $u_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ -4 \end{bmatrix}$, $u_4 = \begin{bmatrix} 5 \\ -3 \\ -1 \\ 1 \end{bmatrix}$, $y = \begin{bmatrix} 10 \\ -8 \\ 2 \\ 0 \end{bmatrix}$.

   (a) Show that $\{u_1, u_2, u_3, u_4\}$ is an orthgonal basis for $\mathbf{R^4}$.

   (b) Suppose $W = \text{Span}\{u_1, u_2, u_3\}$. Why is the set $\{u_1, u_2, u_3\}$ an orthogonal basis of W?

   (c) Write y as the sum of two vectors, one in $W = \text{Span}\{u_1, u_2, u_3\}$ and the other in $W^{\perp}$.

   (d) Let $A = \begin{bmatrix} 0 & 1 & -4 & -1 \\ 3 & 5 & 1 & 1 \\ 1 & 0 & 1 & -4 \end{bmatrix}$. Give a basis for Nul$A$ without any calculation.

Answers are on page 2.

#1(a): Check!

#1(b): If S is an orthogonal set of nonzero vectors, how about a subset of S?

#1(c): Note that $y = a + b$ where $a \in W$ and $b \in W^{\perp}$, and $b = cu_4$ (why?). What is c?

Ans: $y = \begin{bmatrix} 0 \\ -2 \\ 4 \\ -2 \end{bmatrix} + \begin{bmatrix} 10 \\ -6 \\ -2 \\ 2 \end{bmatrix}$.

#1(d): What is Row$A$? Ans: $\{u_4\}$, (or any nonzero scalar multiple).